Compact Evidence Fusion for Reliable LLM Candidate Selection

DASE7506 Advanced Machine Learning Methods - Mini-project 2

Abstract

We study whether comparing candidate solutions improves selection of a correct final numeric answer under a 16 MiB inference-asset and 30-second CPU budget. The selector combines a shared linear embedding scorer, a five-seed relative- embedding MLP ensemble, and shallow boosted trees on candidate features. Component scores are separately normalized and mixed after masking unparseable final answers. The frozen method achieves 384/467 (82.23%) on the complete test bank, compared with 323/467 (69.16%) for final-answer majority voting. It uses 5.127 MiB of extra assets; complete local test prediction plus independent scoring takes 2.150-2.187 seconds across three fresh runs. Validation ablations support modest benefits from explicit consensus features, while a limited arithmetic matcher does not help. Confidence helps order errors but does not establish calibrated reasoning reliability.

1. Research question and task

Can comparing four candidate solutions improve selection beyond judging each candidate independently? Our hypothesis is that agreement and relative candidate information complement learned plausibility evidence, while majority agreement alone can amplify shared mistakes. A second hypothesis is that local arithmetic consistency helps distinguish convincing wrong solutions. We examine both using implemented controls and feature ablations.

The program returns exactly one index in 0..3 per question. The unchanged official scorer compares that candidate's final numeric answer with the reference, including equivalent fractions and decimals. Several candidates may be correct. Correct selection is not proof that every reasoning step is valid; this distinction governs our interpretation of all experimental evidence.

2. Data and development protocol

The supplied GSM8K-derived bank contains 4,755 training, 798 validation and 467 test questions, with four candidate texts and fixed float32 representations of shape [N, 4, 1024] prepared with Qwen3-0.6B. The splits do not overlap. Each question has at least one candidate with a correct final answer. Fitting, normalization and tree-threshold quantiles use training only. Validation guides checkpoint and method selection. Test references are reserved for the frozen method's final scoring and reproduction. There are no external data, additional pretrained weights, online LLM calls or question-ID answer lookups.

Bank construction matters: training and validation include easier groups, whereas test questions have at least two distinct valid parsed final answers. We define **hard validation** by that same input-only rule: at least two distinct non-None outputs of the official parser. It contains 355 of 798 questions; 443 form the complementary easier subset. Majority voting is correct on all 443 easier questions. Thus mixed full-validation accuracy alone hides much of the remaining difficulty. A validation-to-test gap also reflects this construction shift and cannot by itself identify overfitting.

Training observations motivate controls. Of 4,755 training groups, 2,242 have four correct candidates. Candidate positions have similar correctness rates (75.04%-75.79%). Correct candidates average 464.8 characters versus 527.4 for incorrect candidates, but the shortest-answer control performs poorly. Correlations in presentation are diagnostics, not evidence that length is a sound substitute for interpreting the problem.

3. Models and selection procedure

3.1 Training-free controls

The first-candidate control returns index 0. Majority voting groups the official parsed numeric answers, gives invalid answers zero votes, and selects the first member of the largest group. Majority-shortest uses length to break voting ties; shortest ignores votes. These controls expose the value of agreement and the weakness of a simple style heuristic.

3.2 Shared embedding scorers

The linear model shares one scorer across 1024-dimensional candidate vectors. The nonlinear model concatenates each vector with its difference from the within-question mean, yielding 2048 features, and uses one 128-unit ReLU hidden layer. It is trained on hard training groups. Both standardize using training- only means and scales, with a minimum scale of 0.05.

The objective is negative log softmax probability mass on the set of correct candidates, plus 0.25 times candidate-level binary cross entropy. A positive-set loss avoids arbitrarily designating only one correct candidate as positive. Adam uses batches of 128, at most 70 epochs and patience 12. The linear scorer uses seed 17, learning rate 0.002 and L2 0.001, retaining epoch 31. The MLP uses learning rate 0.001, L2 0.005 and dropout 0.3. Its initial seed 29 retains epoch 16; subsequent seeds are 31, 37, 41 and 43. All five predeclared seeds are retained equally, without choosing or discarding lucky seeds. Their raw logits are averaged before the component softmax.

Validation selects checkpoints. Initial fits prioritize full-validation accuracy; hard-trained MLPs prioritize hard-validation accuracy, then full accuracy and positive-set loss. Neither gradients nor normalization fit validation labels. Direct embedding rows in Table 1 use raw logits; fusion additionally masks invalid parsed final answers.

3.3 Text and candidate-relative features

The text module constructs 96 raw features and their 96 within-question centered counterparts. Groups cover length/format and lexical style, question overlap, numeric-answer properties, simple arithmetic equalities and agreement among candidate answers. It uses neither IDs nor cached predictions. Arithmetic expressions use an AST whitelist of numeric constants and addition, subtraction, multiplication and division; Python eval is not used.

A deterministic NumPy histogram gradient-boosting implementation learns scores with listwise softmax cross entropy against a uniform target over correct candidates. Quantile thresholds are fitted on training only. The common ablation recipe is depth 3, 32 bins, learning rate 0.06, L2 10, minimum leaf count 50, at most 140 trees and validation selection every 10 iterations. Seed 2026 is recorded although these tree runs are deterministic. The retained arithmetic-free model uses 180 features and 110 trees.

3.4 Frozen fusion

The linear scorer, mean MLP logits and text scores are converted separately to masked softmax distributions at temperature 1. Invalid parsed final answers are excluded before normalization. Their fixed mixture weights are 0.125, 0.125 and 0.75; the largest mixed candidate score is selected.

Two component combinations were initially searched. Each used the same predeclared grid of text weights {0, 0.25, 0.5, 0.75, 1}, answer-group sum-pooling weights {0, 0.5, 1} and explicit vote weights {0, 0.1, 0.25}: 45 configurations per round, 90 total. Both selected zero extra pooling and zero extra votes. The five-seed extension keeps those scalar settings unchanged. Zero added pooling is not evidence that cross-candidate information is useless: the text features and relative embeddings already contain it.

4. Validation comparisons and ablations

Table 1 uses root-run, unchanged official prediction and independent scoring. Full validation contains 798 questions and hard validation 355. Complete time sums fresh prediction-process and scoring-process wall times, including startup, imports, loading, preprocessing and all members. Single-run development times are local observations; the final row shows the range of three fresh runs.

Method	Full accuracy (%)	Hard accuracy (%)	Complete time (s)	Assets (MiB)
First candidate	78.20	54.93	0.836	<0.001
Shortest candidate	69.30	37.75	0.822	<0.001
Majority + shortest tie	84.46	65.07	0.854	<0.001
Final-answer majority	86.22	69.01	0.848	0.000
Linear embedding	91.48	81.69	1.158	0.013

Method	Full accuracy (%)	Hard accuracy (%)	Complete time (s)	Assets (MiB)
Relative MLP, seed 29	90.85	82.54	0.953	1.018
Text trees, full features	90.85	79.44	2.770	0.016
Text trees, no arithmetic	91.35	80.56	2.594	0.022
Text trees, no consensus	90.10	77.75	2.673	0.027
Masked embedding fusion	92.36	82.82	0.993	1.032
Mixed fusion, one MLP seed	93.86	86.20	2.646	1.053
Frozen fusion, five seeds	94.11	86.76	3.122-3.226	5.127

The final model answers 751/798 validation and 308/355 hard questions correctly. Relative to majority it gains 63 questions, or 7.89 full-validation percentage points and 17.75 hard-subset points. Paired outcomes are 674 both correct, 77 only fusion correct, 14 only majority correct and 33 both wrong. A paired bootstrap interval for the full-validation gain is [5.76, 10.28] percentage points. It is conditional on the chosen model and reused validation bank, not corrected for search and not an independent generalization guarantee.

The mixed model improves over its individual components, consistent with complementary errors. Averaging all five predeclared MLP seeds adds two correct validation questions over the single-seed mixture, at approximately 4.07 MiB extra cost. This small gain is weak evidence of a broadly repeatable advantage; we retain it as a fixed stability choice within the resource budget, not as proof that larger ensembles are generally preferable.

The three text-tree variants use the same recipe and search budget. Early stopping selects 80 trees for full features, 110 without arithmetic and 140 without consensus, with earliest checkpoint retained on ties. Removing arithmetic gains four full-validation questions; removing consensus loses six relative to full features. The former rejects the usefulness of this limited arithmetic matcher here, not arithmetic verification generally. It misses units, algebra, nested notation and the interpretation of a correct formula. The latter supports a modest contribution from the explicit consensus group, but other features retain indirect group information. It is not a pure independent-candidate comparison, and selected capacity differs under the shared stopping rule.

The shortest controls show that the training length correlation does not make shorter answers reliably correct. They do not show that learned scorers are free of style shortcuts. A question-plus-answer-only representation comparison would be needed to isolate reasoning text: the supplied embeddings themselves encode candidate traces.

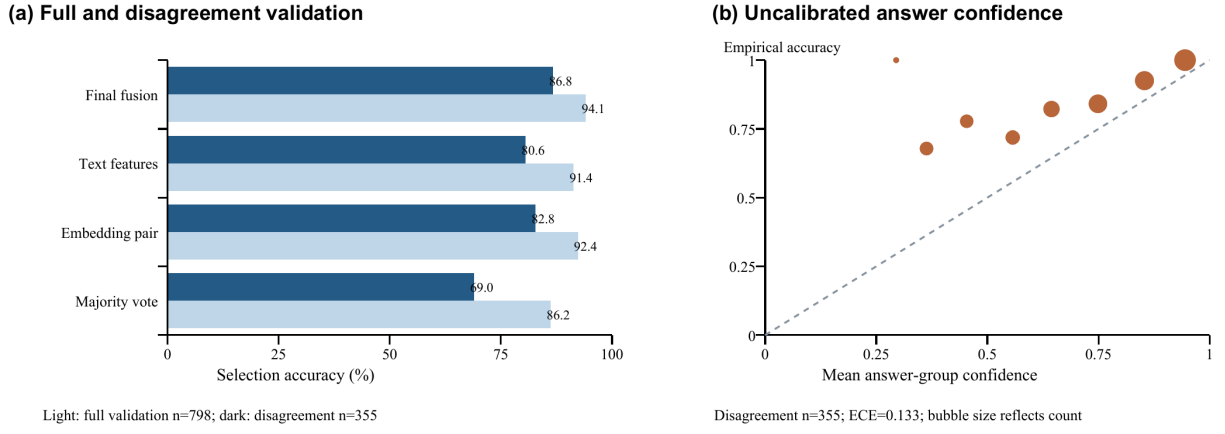


Figure 1. Left: full-validation and input-defined hard-validation accuracy. Right: hard-validation confidence-bin correctness versus mean selected-answer probability mass. Confidence has not been calibrated. ECE is a descriptive statistic on the same validation data used for model selection; it does not establish out-of-sample calibration or reasoning validity.

5. Confidence and failure analysis

Diagnostic confidence is the mixed probability mass assigned to candidates sharing the selected final answer; it is not automatically a calibrated correctness probability. Ten-bin expected calibration error is 0.0591 and Brier score 0.0578 on full validation, versus 0.1330 and 0.1298 on hard validation. The easier groups inflate apparent reliability: 535 questions have confidence at least 0.9, but only 92 are hard. Those 535 choices are all correct in this validation sample, not a promise of error-free high-confidence predictions.

Ranking by confidence gives 200/200 correct at 25.06% coverage, 399/399 at 50%, and 594/599 (99.17%) at 75.06%, compared with 751/798 (94.11%) at full coverage. These are retrospective diagnostics; the task requires an answer for every question, so inference does not abstain. Reused validation data, small low-confidence bins and easy-subset concentration limit calibration and selective-prediction claims.

The saved validation errors illustrate semantic weaknesses. In the naps question, three two-hour naps per week over 70 days should total 60 hours, but a selected trace multiplies 3*2*70 to obtain 420. Every multiplication is numerically correct; the missing step is conversion from days to weeks, and both answers have two votes. In the stock question, a candidate returns gross receipts of 100 instead of subtracting the original cost to reach profit 40. In the student-pairing question, repeated answers can share an incorrect counting interpretation. The candidate with final value 2 also contains flawed intermediate statements about totals and pair counts. These examples show why local arithmetic, consensus and fluent presentation do not establish semantic correctness, and why an accepted final number need not imply a valid explanation.

6. Frozen test result and resources

The release was frozen at 2026-10-22 08:11:43 UTC as mp2-751aa13c6e1c. The freeze manifest records inference-source hashes and all matching asset hashes and predeclares only the final selector and majority baseline for test scoring. No method or setting is changed in response to test results.

Frozen test method	Correct / questions	Accuracy (%)
Final-answer majority	323 / 467	69.1649
Final five-seed fusion	384 / 467	82.2270

The test gain is 61 questions, or 13.0621 percentage points. All test groups are hard by construction, so the appropriate difficulty-matched validation comparison is 86.76% on hard validation versus 82.23% on test, rather than 94.11% on the mixed validation bank. Distribution differences and validation selection optimism can both contribute; neither can be ruled out by these figures. This evaluation concerns the published course bank, not arbitrary mathematical reasoning.

The final bundle has 5,375,667 counted uncompressed bytes (5.127 MiB), including six embedding checkpoints, one tree checkpoint and configuration. Prediction plus independent scoring takes 2.1535, 2.1496 and 2.1866 seconds in three fresh local test runs; validation takes 3.1977, 3.1220 and 3.2262 seconds. Timings include startup, imports, assets/data loading, preprocessing and every member. Eligibility is decided on the reference Xeon Platinum 8457C, not these local measurements.

Development used CPU only, Python 3.12.14 and NumPy 2.5.3 on Windows 11 with 12 logical CPUs and 15.73 GiB RAM. The processor identifier is Intel64 Family 6 Model 154 Stepping 3, GenuineIntel. Training used one OpenBLAS/OpenMP/MKL thread; official local inference configures four total threads. No GPU was used.

The recorded search comprised 18 embedding configurations, four centered-ridge regularizers, five text configurations, 90 scalar fusion-grid settings and one fixed five-seed comparison. Recorded cost sums are 77.577 seconds for embedding training, 1.469 seconds for ridge fitting plus shared preprocessing, 195.214 seconds for text experiments, and 6.741 seconds for fusion search and fixed rebuilding. These timers have different scopes; some text experiments reuse deterministic input-only feature caches. They are not presented as an exact total orchestration wall time. The tracked resource summary gives each count and timing scope.

A fresh complete reconstruction of the frozen seven models and configuration, including new text-feature extraction, took 80.315 seconds in the recorded builder. All seven reconstructed weight files match the frozen SHA-256 hashes. Independent reconstruction acceptance also confirms semantic configuration equality and identical choices on all 798 validation questions. The tracked evidence directory records the frozen recipe, seeds,

selected checkpoints, exact commands, external timings and asset hashes. Checkpoint scoring requires no retraining. Root acceptance separately covers new IDs, different bank sizes, singleton/reordered inputs, candidate permutation, input non-mutation and inference reads limited to the checkpoint; these checks are not additional training or test-guided model selection.

Quantitative sources are the tracked evidence/official records, evidence/validation_analysis.json, evidence/resource_summary.json and evidence/reproduction_acceptance.json in code.

7. Limitations and conclusion

Repeated validation-based architecture, checkpoint and mixture selection introduces optimism. The reported bootstrap and confidence diagnostics do not remove it; hard validation is also reused rather than an independent holdout. Fixed references may not capture every ambiguity in natural-language questions. Correctness here follows the official reference and parser, not an independent audit of each derivation.

The evidence supports a compact combination of learned plausibility and candidate-relative information for this bank. Explicit consensus features help modestly under a controlled recipe; simple arithmetic matching does not. The final offline fusion improves frozen-test accuracy while using about 5.13 MiB and a small local CPU budget. Its selections should still be understood as estimates of final-answer correctness, not certificates of sound reasoning.

8. AI assistance and attribution

ChatGPT/Codex provided substantive multi-agent assistance with interpreting requirements, implementing models and training, developing experimental tools, analysing validation results and drafting this report and README. A coordinating agent performs independent official evaluation and integration. The student is responsible for reviewing and understanding the implementation and claims; this must not be represented as unaided student work.

Questions and references derive from GSM8K under its retained MIT license. Candidate solutions and fixed representations were prepared with Qwen3-0.6B. No additional pretrained weights are used. The supplied license and attribution files are retained; Python's standard library and NumPy support student code.

References

[1] DASE7506 MP2 guide. https://github.com/xudongwu-0/DASE7506-MP2/blob/main/guide/GUIDE.md

[2] DASE7506 MP2 running instructions. https://github.com/xudongwu-0/DASE7506-MP2/blob/main/code/README.md

[3] GSM8K. https://github.com/openai/grade-school-math

[4] Qwen3-0.6B attribution. https://huggingface.co/Qwen/Qwen3-0.6B